

RAJALAKSHMI ENGINEERING COLLEGE

An Autonomous Institution, Affiliated to Anna University

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Web Technology And Mobile Application

Experiment 8

2026-2027

By

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AIM:

To develop a **web-based Online Railway Reservation System** that validates user input using JavaScript and performs **server-side processing using Servlet**, storing reservation details in a database using JDBC.

ALGORITHM:

1. Create a **Dynamic Web Project** in Eclipse.
2. Design a **JSP page (exp8.jsp)** with a reservation form.
3. Implement **JavaScript validation** to check required fields and valid inputs.
4. Create a **Servlet (RailwayReservationServlet.java)** to handle form submission.
5. Perform **server-side validation** in the servlet.
6. Establish **database connectivity using JDBC**.
7. Create a **MySQL database and reservations table**.
8. Insert user data into the database using PreparedStatement.
9. Display a **styled success/failure response page**.
10. Deploy the project on Apache Tomcat server and run using localhost.

CODE:

HTML:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Library Management - Issue Book</title>
<style>
body { font-family: sans-serif; margin: 40px; line-height: 1.6; }
.container { max-width: 400px; border: 1px solid #ccc; padding: 20px; border-
radius: 8px; }
input { width: 100%; padding: 8px; margin: 10px 0; display: block; }
button { background: #007bff; color: white; border: none; padding: 10px; cursor:
pointer; width: 100%; }
#errorLog { color: #d9534f; font-size: 0.9em; margin-bottom: 10px; }
</style>
</head>
<body>

<div class="container">
<h2>Issue Book</h2>
<div id="errorLog"></div>
<form id="LibForm" action="LibraryServlet" method="POST">
```

```
<label>Member ID:</label>
<input type="text" id="memberId" name="memberId" placeholder="LIB-1234">
<label>Book ISBN:</label>
<input type="text" id="isbn" name="isbn" placeholder="10 or 13 digits">
<label>Return Date:</label>
<input type="date" id="returnDate" name="returnDate">
<button type="submit">Update Database</button>
</form>
</div>
```

```
<script>
document.getElementById('libForm').onsubmit = function(e) {
const memberId = document.getElementById('memberId').value;
const isbn = document.getElementById('isbn').value;
const returnDate = new Date(document.getElementById('returnDate').value);
const today = new Date();
const errorLog = document.getElementById('errorLog');
let errors = [];
```

```
// Validation: Member ID Format
if (!/^LIB-\d{4}$/.test(memberId)) {
errors.push("Invalid Member ID. Format: LIB-XXXX");
}
```

```
// Validation: ISBN Length
if (isbn.length !== 10 && isbn.length !== 13) {
errors.push("ISBN must be 10 or 13 digits.");
}
```

```
// Validation: Date Logic
if (isNaN(returnDate.getTime()) || returnDate <= today) {
errors.push("Return date must be in the future.");
}
```

```
if (errors.length > 0) {
e.preventDefault(); // Stop form submission
errorLog.innerHTML = errors.join("<br>");
return false;
}
return true;
};
</script>
```

```
</body>
</html>
```

Servlet code:

```
package wtma;
import java.io.*;
import java.sql.*;
import jakarta.servlet.*;
import jakarta.servlet.http.*;
import jakarta.servlet.annotation.*;

@WebServlet("/LibraryServlet")
public class exp8 extends HttpServlet {

    private static final long serialVersionUID = 1L;
    private final String DB_URL = "jdbc:mysql://localhost:3306/library_db";
    private final String USER = "root";
    private final String PASS = "password123";

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String memberId = request.getParameter("memberId");
        String isbn = request.getParameter("isbn");
        String returnDate = request.getParameter("returnDate");

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection conn = DriverManager.getConnection(DB_URL, USER, PASS);

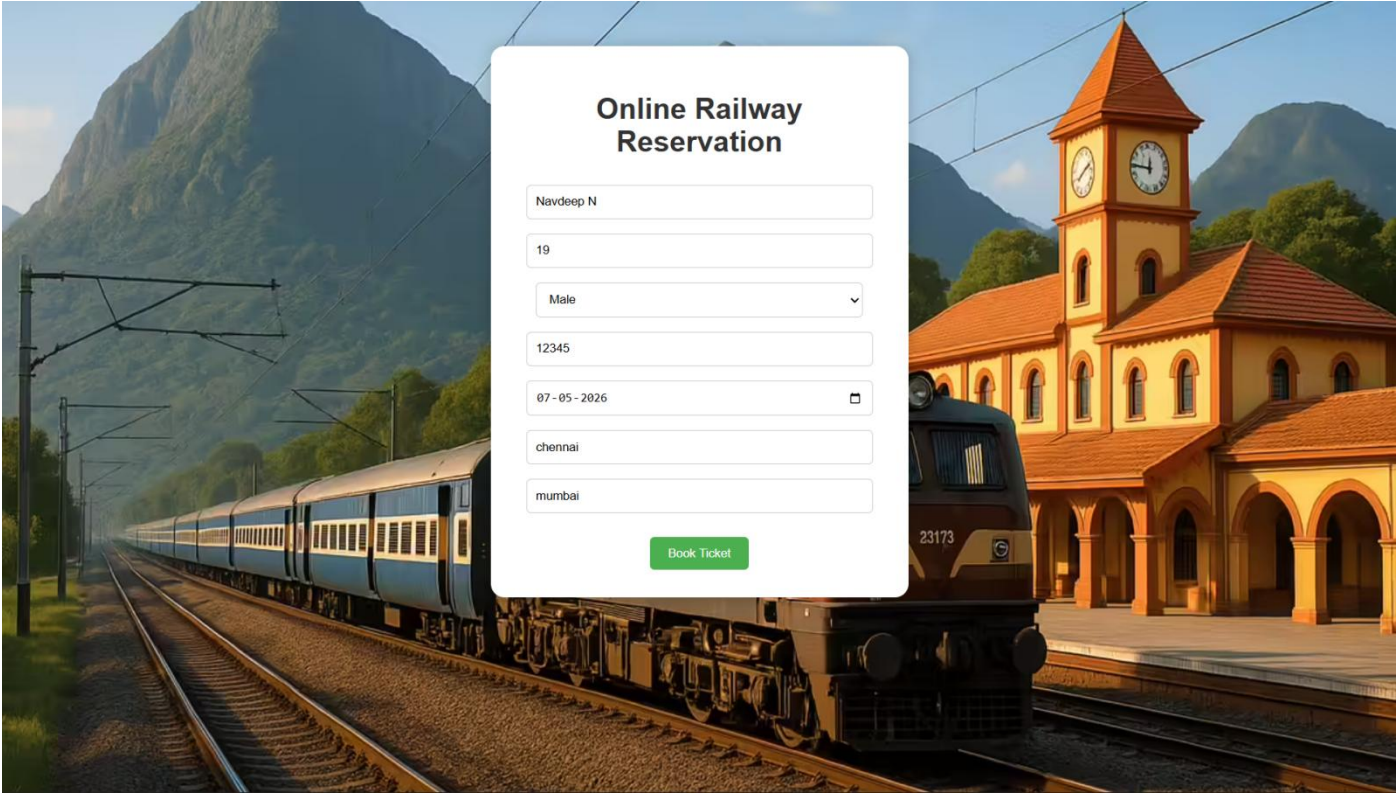
            String sql = "INSERT INTO book_issues (member_id, isbn, return_due_date) VALUES (?,?, ?)";
            PreparedStatement pstmt = conn.prepareStatement(sql);
            pstmt.setString(1, memberId);
            pstmt.setString(2, isbn);
            pstmt.setString(3, returnDate);

            int rows = pstmt.executeUpdate();

            if (rows > 0) {
                out.println("<h3>Success! Book issue recorded in database.</h3>");
                out.println("<a href='index.html'>Go Back</a>");
            }

            conn.close();
        } catch (Exception e) {
            out.println("<h3>Database Error:</h3> " + e.getMessage());
            e.printStackTrace(out);
        }
    }
}
```

OUTPUT:



Online Railway Reservation SystemAdmin Dashboard ▾

System Status and Updates

✓ Success! Booking reference table has been updated.

Update timestamp: 2023-11-15 14:30:00 PST

Current Booking Period: LIB-1234 (Active)
Total Records Updated: 100,456
System Status: OPTIMAL
Last Full Backup: 2023-11-14 23:59:00 PST

Refresh System Status

Recent System Activities

Time	Activity	Status
2023-11-15 14:30:00	Table Update:	Success
2023-11-15 13:45:12	Cache Clear:	Partial
2023-11-15 12:00:05	User Audit:	Success

© 2023 Rail Reservation Solutions Inc. | System Status: Optimal

RESULT:

The Online Railway Reservation System was successfully implemented. User inputs were validated on both client and server sides, and reservation details were **stored in the MySQL database**. A **styled output page** displayed the reservation status correctly.